

Honeywell

LCNP Status User's Guide

EP-DSX305

EX24-240

R240

03/08

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Release 240 March 2008**

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About This Document

This document describes the LCNP Status display that is the "control panel" for the TotalPlant Solution (TPS) System portion of your station. The LCNP status is available on Experion and TPS nodes using the LCNP/LCNP4/LCNP4M boards.

Release Information

Document Name	Document ID	Release Number	Publication Date
LCNP Status User's Guide - EX24	EX24-240	240	07/05
LCNP Status User's Guide - EX24 EP-DSX305	EX24-240	240	03/08

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Document ID
Native Window User's Guide	GU20
Configuration Utility	TP20W

Contacts

World Wide Web

The following Honeywell web sites may be of interest to Process Solutions customers.

Honeywell Organization	WWW Address (URL)
Corporate	http://www.honeywell.com
Honeywell Process Solutions	http://hpsweb.honeywell.com

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.

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1. Determining LCNP Status

1.1 LCNP Status Introduction

About the LCNP status dialog

The LCNP Status Dialog is the "control panel" for the TotalPlant Solution (TPS) System portion of your workstation. It provides information previously available in physical LEDs and other alphanumeric board indicators.

The LCNP status is available on Experion and TPS nodes using the LCNP/LCNP4/LCNP4M boards. This includes the following node types:

- Global User Station (GUS)
- Application Processing Platform (APP)
- LCN-connected ES-C
- LCN-connected Server
- LCN-connected ACE

Relationship to Native Window

The LCNP Status Dialog functions independently of the Native Window. Although much of the data displayed may be found among various Universal Station displays, additional information is presented that is not readily available elsewhere.

Modes of operation

The LCNP Status Dialog runs in three modes:

- Normal (No Details)
- Extended (Details)
- Iconized



REFERENCE - INTERNAL

LCNP Status can also be connected remotely. Refer to "Native Window User's Guide" topic **Remoting** for details.

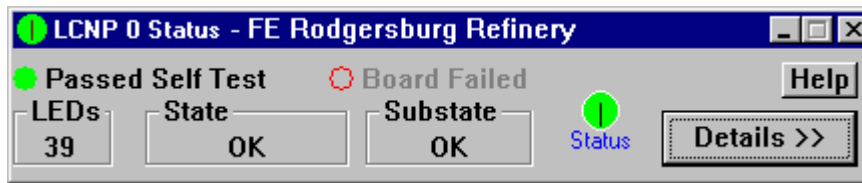
1. Determining LCNP Status

1.2. Interpreting Normal Mode Indicators

1.2 Interpreting Normal Mode Indicators

Invoking the LCNP status

Select the LCNP Status menu selection from the Start menu. The LCNP Status dialog opens in normal (No Details) mode, as pictured below.



Normal mode information

Normal mode provides the following critical information for monitoring the overall status of the LCNP:

- Passed Self Test indication
- Board Failed indication
- LEDs values
- State
- Substate
- Status indication
- Help
- Details

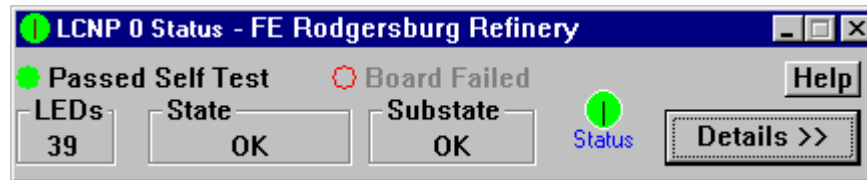
About normal mode indicators

When using the status in normal mode, you typically view the state and status indications to determine the node's operational and loading states.

Passed self test indicator

Passed Self Test is an indicator of the self-test status of this LCNP board.

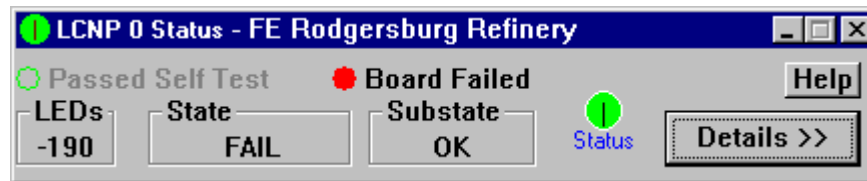
The hollow green circle fills solid green, as pictured below, if the LCNP board has successfully passed self-test.



Board failed indicator

Board Failed is an indicator of the runtime status of this LCNP.

The hollow red circle fills solid red, as pictured below, if the LCNP has failed.



LEDs

The LEDs field represents the LCN address of the node. The LEDs field number is previously defined from the Configuration Utility Board dialog to match the address number defined in the NCF.

This data is equivalent to the K2LCN/K4LCN board alphanumeric LEDs in classic TPS nodes. During the power-up/self-test sequence, the values displayed denote progress through self-test. If the self-test program completes without failure, the configured LCN node address is displayed. If a failure occurs during self-test or normal operation, an error identification code is displayed.

1. Determining LCNP Status

1.2. Interpreting Normal Mode Indicators

State

This data reflects the current TPS node state.

When the State indicator reads	The node
NOT CONFIG	is not configured
OFF	is powered off
PWR_ON	is powered on and awaiting load
LOC LOAD	is loading from a local source
NET LOAD	is loading over the LCN network
TEST	is under test
QUALIF	has passed self-test
READY	has completed loading and is in initialization
OK	is running normally
TERM	is terminating
FAIL	has failed

Substate

This data reflects the current TPS node substate.

When the Substate indicator reads	The node
OK	is running normally
WARNING	has experienced minor difficulty loading
SEVERE	has experienced major difficulty loading
FAILED	has failed
ISOLATED	is isolated from the network

Status applet

The vane in the Status Applet moves at the refresh rate specified to illustrate that the display is functioning properly.

The LCNP Status Applet's color indicates status as follows:

If the Status Applet color is	TPS node is in one of the following states
Red	FAIL TERM OFF NOT CONFIG
Yellow	QUAL PWR ON LOC LOAD NET LOAD TEST READY
Green	OK

If the Status Applet is connected REMOTELY to an LCNP, it should not be set at update rates faster than once per second except for diagnostic purposes. Doing so will adversely affect the update rates achievable by the Native Window.

1. Determining LCNP Status

1.2. Interpreting Normal Mode Indicators

Help button

The **Help** button calls up the on-line help documentation.



Details>> button (from normal mode)

You will usually be operating in the normal mode, but on occasion will need to access the details to:

- Get substate information for Honeywell field technicians
- Reset the LCNP
- Specify optional time set settings

Click the **Details>>** button to expand the LCNP Status dialog to the extended mode.



The **Details>>** button becomes the **No Details** button.



1.3 Interpreting Extended Mode Indicators

Expanding LCNP status

Click **Details>>** on the normal mode LCNP Status display. The LCNP Status dialog expands to extended mode, as pictured below.

The screenshot shows a Windows-style dialog box titled "LCNP 0 Status - FE Rodgersburg Refinery". At the top, there are two status indicators: a green circle with a checkmark labeled "Passed Self Test" and a red circle with an 'X' labeled "Board Failed". Below these are three fields: "LEDs" with the value "39", "State" with "OK", and "Substate" with "OK". To the right of these fields is a green circle with a downward arrow labeled "Status" and a "No Details" button. A "Help" button is in the top right corner. The main area of the dialog is divided into several sections. The "LED Description" section has a "Node Address" field. Below this are three sections for different slots: "Slot 2: WSI2", "Slot 3: SCSI", and "Slot 4: EPDG". Each slot section contains a row of hexadecimal values followed by two status indicators (a green circle with a checkmark and a red circle with an 'X'). The "Bar LED Status" section contains nine status indicators arranged in a 3x3 grid, each with a red circle and an 'X'. The "Refresh" section has a "Refresh Now" button, a "Refresh" label, a numeric input field with the value "1", two radio buttons labeled "Seconds" and "1/10 Second", and a "Reset LCNP" button. The "Revisions" section contains a single line of text: "K2LCN_8MW: 4C/0C K2LCN_LCN: 2G/0P Serial: 1 Driver: 1".

Node Address	Slot 2: WSI2	Slot 3: SCSI	Slot 4: EPDG
4039 001c 5555 aaaa 5555 aaaa 0000 009d	47ca 0702 0000 0099 0280 8e00 3f3f 0101	4742 1002 6056 30a1 0280 8420 0f00 0101	

Bar LED Status		
Self Test	Data Compare Error	Data Parity Error
Bus Error	Data Ack Timeout	Grant Timeout
No Data	Single Bit Error	Multiple Bit Error

Refresh: ☒ Seconds ☐ 1/10 Second

Revisions: K2LCN_8MW: 4C/0C K2LCN_LCN: 2G/0P Serial: 1 Driver: 1

1. Determining LCNP Status

1.3. Interpreting Extended Mode Indicators

Extended mode information

Extended mode provides additional hardware and diagnostic information, such as:

- LED descriptions
- Slot designators
- TPS slot assignments
- Slot register contents
- Various LED indicators
- LCNP board hardware/firmware revision information

Extended mode is intended as an aid to the field technician.

No Details button (from extended mode)

Click **No Details** to return the LCNP Status dialog to the normal mode of operation. The **No Details** button becomes the **Details>>** button.

LED description

LED Descriptions explain the meaning of codes that appear in the LEDs field.

Slot designators

These labels reflect the names of the boards being emulated on a slot-by-slot basis.

Board selection and activation is specified via the Configuration Utility.

Slot # assignments

Three slots can be assigned to contain emulated TPS boards of the following types:

- WSI - Work Station Interface board
This emulator supports X-layer functions.
- SCSI - Disk Drive board
This emulator supports emulated disks.
- PDG - Human Interface board
This emulator supports video, keyboard, mouse, and touchscreen.

TPS boards are also referred to by their slot numbers.

Slot register contents

This data reflects the slot register contents for each of the emulated slots. During operations, the register contents change at a very high rate, but the data is updated only at the designated refresh rate. Slot Register data is of a diagnostic nature and is normally of use only to Honeywell Field Service personnel.

Slot passed-self-test indicators

Each emulated slot has a self-test status indicator.

The hollow green circle fills solid green, as pictured below, if the emulator slot has successfully passed self-test.

The screenshot shows a software window titled "LCNP 0 Status - FE Rodgersburg Refinery". At the top, it displays "Passed Self Test" with a solid green circle and "Board Failed" with a hollow red circle. Below this, there are fields for "LEDs" (39), "State" (OK), and "Substate" (OK). A "Status" button with a green circle and a "No Details" button are also present. The main section lists slot information:

Slot	Description	Node	Address	Value 1	Value 2	Value 3	Value 4	Value 5	Value 6	Value 7	Value 8	Value 9	Value 10	Self Test	Board Failed
Slot 2:	WSI2	4039	001c	5555	aaaa	5555	aaaa	0000	009d					Solid Green	Hollow Red
Slot 3:	SCSI	47ca	0702	0000	0099	0280	8e00	3f3f	0101					Solid Green	Hollow Red
Slot 4:	EPDG	4742	1002	6056	30a1	0280	8420	0f00	0101					Solid Green	Hollow Red

Below the slot information is a "Bar LED Status" section with a grid of indicators, all of which are currently hollow red circles:

Self Test	Data Compare Error	Data Parity Error
Bus Error	Data Ack Timeout	Grant Timeout
No Data	Single Bit Error	Multiple Bit Error

At the bottom, there is a "Refresh" section with a "Refresh Now" button, a "Refresh" dropdown set to "1" with options for "Seconds" (selected) and "1/10 Second", and a "Reset LCNP" button. The "Revisions" section at the very bottom shows: "K2LCN_8MW: 4C/0C K2LCN_LCN: 2G/0P Serial: 1 Driver: 1".

If neither circle is solid, the corresponding slot emulation is disabled.

1. Determining LCNP Status

1.3. Interpreting Extended Mode Indicators



ATTENTION

After reset and self-test, the WSI2 Slot self-test indicators are not set. They become set only after the initial load sequence is complete and QLT tests have evaluated the board for proper functioning.

Slot failed indicators

Slot Failed is an indicator of the runtime status of a board in a slot.

The hollow red circle fills solid red, as pictured below, if the emulator slot has failed.

LCNP 0 Status - FE Rodgersburg Refinery

Passed Self Test **Board Failed** **Help**

LEDs: 39 State: OK Substate: OK Status: No Details

LED Description

Node Address

Slot 2: WSI2
4039 001c 5555 aaaa 5555 aaaa 0000 009d ● ●

Slot 3: SCSI
47ca 0702 0000 0099 0280 8e00 3f3f 0101 ● ○

Slot 4: EPDG
4742 1002 6056 30a1 0280 8420 0f00 0101 ● ○

Bar LED Status

○ Self Test ○ Data Compare Error ○ Data Parity Error
○ Bus Error ○ Data Ack Timeout ○ Grant Timeout
○ No Data ○ Single Bit Error ○ Multiple Bit Error

Refresh **Refresh Now** ☒ Seconds ☐ 1/10 Second **Reset LCNP**

Revisions
K2LCN_8MW: 4C/0C K2LCN_LCN: 2G/0P Serial: 1 Driver: 1

If neither circle is solid, the corresponding slot emulation is disabled.

Bar LED status

The red circles in the Bar LED Status area are indicators for specific functions and errors on the LCNP board.

Following reset, the red **Self Test** circle fills solid red during the LCNP board self-test, as pictured below, and then returns to a hollow circle.

LCNP 0 Status - FE Rodgersburg Refinery

☒ Passed Self Test
 ☒ Board Failed
 ☒ Status

LEDs: -13 State: NOT CONFIG Substate: OK

LED Description

in RAM test

Slot 2: WSI2?

0039	0008	5555	aaaa	5555	aaaa	0000	0008	☒	☒
------	------	------	------	------	------	------	------	----------------------------------	----------------------------------

Slot 3: SCSI?

07ca	0700	0000	0098	0e00	4000	aaaa	5555	☒	☒
------	------	------	------	------	------	------	------	----------------------------------	----------------------------------

Slot 4: EPDG?

0742	1000	0000	30a1	0200	8420	aaaa	5555	☒	☒
------	------	------	------	------	------	------	------	----------------------------------	----------------------------------

Bar LED Status

☒ Self Test	☐ Data Compare Error	☐ Data Parity Error
☐ Bus Error	☐ Data Ack Timeout	☐ Grant Timeout
☐ No Data	☐ Single Bit Error	☐ Multiple Bit Error

Refresh: ☒ Seconds ☐ 1/10 Second

Revisions

K2LCN_8MW: 4C/0C K2LCN_LCN: 2G/0P Serial: 1 Driver: 1

The hollow red circles fill solid red if there is a problem in any of these areas.

1. Determining LCNP Status

1.3. Interpreting Extended Mode Indicators

Refresh Now button

The **Refresh Now** button causes a manual refresh of all data displayed in the LCNP Status dialog.



Refresh interval

This control allows the user to select the rate at which the LCNP Status dialog refreshes its data.

One of two time increments can be specified:

- Seconds
A rate of one second provides a reasonable balance between freshness of data and overhead to the system.
- Tenths of seconds
The maximum rate of 1/10 second is useful only in certain diagnostic situations and is not recommended during normal operation.

Setting the update rate to zero inhibits automatic update. In this case, the **Refresh Now** button must be used to update data.

The vane in the Status icon moves at the rate specified to illustrate that the display is functioning properly.

Reset LCNP button

This button performs a RESET on the LCNP, the same function initiated by pressing the red reset button located under the top edge of the Operator keyboard on a classic Universal Station or the white reset button on a US electronics chassis.

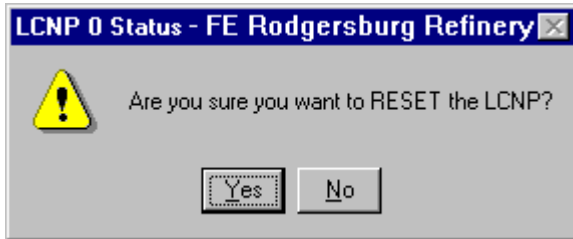


WARNING

Resetting an LCNP interrupts all processing on the LCNP and causes loss of process view in operator stations. The node must be RELOADED with its personality prior to further use.

Resetting an LCNP

Use the following procedure to reset the LCNP.

Step	Action
1	Click Reset LCNP. The following dialog appears:  This provides the opportunity to cancel the reset operation by clicking No.
2	Click Yes to continue. The LCNP now resets.

Reset LCNP is not recommended as a preferred mechanism for shutting down the LCNP. The preferred method is to perform a NODE SHUTDOWN from the Native Window's System Status display whenever possible.

1. Determining LCNP Status

1.3. Interpreting Extended Mode Indicators

Reset LCNP can be configured in Configuration Utility to be active or nonactive for the LCNP Status display. When non-active, the **Reset LCNP** button appears with gray text.

TPN time synchronization

LCNP4-GPS boards support a TPN time synchronization option that allows you to synchronize TPN time to TPS time. An LCNP4-GPS board displays buttons in the details mode to adjust the time synchronization. A node can leverage this time sync capability only when configured as a Clock Source in the NCF. Other GPS-enabled nodes become listener nodes. Once configured as a Clock Source, enable and configure time sync on the clock source node.



REFERENCE - INTERNAL

The time synchronization configuration and operation are described in TP20W Configuration Utility.

LCNP 0 Status -

☒ Passed Self Test ☐ Board Failed

LEDs: 90 State: OK Substate: OK Status: ☒ No Details:

LED Description

Node Address	LEDs	Status
Slot 2: WSI2		
4039 001c 0000 0000 0000 0000 0000 00da	☒	☐
Slot 3: SCSI		
47ca 0702 0000 0099 0280 8e00 3f3f 0101	☒	☐
Slot 4: EPDG		
4742 1002 0001 30a1 0280 8420 0000 0000	☒	☐

Bar LED Status

☐ ST ☐ BE ☐ ND ☐ DCE ☐ DAT ☐ SBE ☐ DPE ☐ GT ☐ MBE

Refresh

☒ Seconds ☐ 1/10 Second

Time Sync

TPS Time: 10:49:38 TPN Time: 10:49:33

Last Execution Result: at 10:49:35 [TPS time]
Gradual Adjustment In Effect, TPN time was 10:49:30

Revisions

LCNP4-GPS: 5N/0N LCN_CHIPS: 3K/0C Serial: 2 Driver: 3

Revisions

The current revision identifiers of the hardware/firmware in use are indicated as follows:

LCNP:

Displayed as "AB/CD" where

A = Hardware Version (4)

B = Hardware Revision (B initially)

C = Firmware Version (always 0)

D = Firmware Revision (B for initial version)

of the microprocessor section of the LCNP

LCNI:

Displayed as "AB/CD" where

A = Hardware Version (2)

B = Hardware Revision (G initially)

C = Firmware Version (always 0)

D = Firmware Revision (P initially)

of the LCN Interface section of the LCNP

Serial

Serial is indicated by "xxx" where xxx is the version number of the Serial Port PROM.

Driver

Driver is indicated by "xxx" where x is the version of the workstation Driver software.

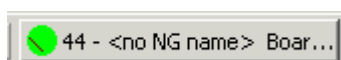
1. Determining LCNP Status

1.4. Iconized Mode

1.4 Iconized Mode

Iconizing the LCNP status

Click the minimize button in the upper-right corner of the LCNP Status dialog. The dialog is reduced to a button on the taskbar located at the bottom of the desktop.



The LCNP Status icon has two parts:

- **Status Applet**
The Status applet shows the LCNP Status display refreshing activity.
- **LEDs value**
The LEDs value displayed below the status applet is the value shown in the LEDs frame of the LCNP Status display dialog. This number corresponds to the physical 3-digit LED indicator found on the classic TPS K2LCN/K4LCN board.

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